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SAMConvFormer+LLM: Exploring synergistic fusion of Segment Anything Model with joint convolutional transformer and large language model to advance dense agricultural crop analysis☆

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ABSTRACT

Accurate detection of densely fruited crops is essential for yield estimation, ripeness analysis, and actionable recommendations for optimized crop management using large language models. However, real-world agricultural environments present significant challenges, including varying illumination, diverse fruit sizes, and highly complex backgrounds, making robust detection non-trivial. Existing AI models in agriculture, customized on controlled datasets, struggle to generalize to real-field conditions, limiting their practical applicability. To address this, we introduce a real-field dataset capturing the complexities of a dense crop, which serves as a benchmark to expose model limitations and drive the development of more robust methods. Leveraging this, we proposed a novel detection framework (SAMConvFormer) that synergizes the Segment Anything Model (SAM) with a Joint Convolutional Transformer, combining their unique strengths for robust agricultural crop detection. This integration of SAM further significantly improves boundary delineation, particularly for overlapping clusters, blurring, and irregular fruit structures. In addition, we integrate the proposed SAMConvFormer framework with a Large Language Model to develop a comprehensive pipeline for yield estimation, ripeness analysis, and actionable recommendations for optimized crop management. The proposed model achieves state-of-the-art performance with 79.32% sensitivity, 62.64% IoU, and 77.03% Dice coefficient, representing relative improvements of 31%, 18.79% and 11.56% over the baseline model, highlighting its robustness and precision for the detection of densely fruited crops. To advance this research, we will publicly release the proposed framework and related materials, which offer a solid foundation for future studies. GitHub repository at https://github.com/Owais-CodeHub/SAMConvFormer-LLM.

1. Introduction

The integration of Large Language Models (LLMs) in agriculture has opened new avenues to address complex challenges in precision farming (Kuska et al., 2024). By leveraging their advanced capabilities in contextual understanding and decision-making, LLMs are transforming key aspects of agricultural management, including yield estimation, ripeness analysis, and actionable recommendations for optimized crop production (Qing et al., 2023). Recent advances in the Artificial Intelligence (AI) era have significantly impacted precision agriculture, enabling novel solutions to complex challenges such as dense crop segmentation (Xie et al., 2021; Liu et al., 2021; Singh et al., 2024).

However, achieving accurate and robust detection of densely fruited crops in real-world agricultural environments remains a significant challenge. Factors such as a lack of real-field datasets that accurately capture the complexities of practical farming environments further limits the effectiveness of AI-based solutions. Unlike controlled settings, real-field conditions present challenges such as variable lighting, occlusions, and densely clustered fruits, making it significantly more difficult to develop models that generalize effectively (Muresan and Oltean, 2018; Yu et al., 2019). In recent years, several studies have focused on dataset creation to improve the performance of fruit segmentation models. For instance, to address the need for specialized datasets, FruitSeg30 (Shamrat et al., 2024) was introduced, offering annotated images to evaluate segmentation models. A Unet-based model (Ronneberger et al., 2015) trained on this dataset demonstrated promising results. However, FruitSeg30 lacks the scale and complexity needed to address real-world agricultural challenges, particularly those involving densely fruited crops with significant occlusions and structural

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variations. To overcome these limitations, researchers have explored synthetic datasets (Gai et al., 2024), which simulate real-world scenarios to expand training data, and weakly supervised learning methods (Kong et al., 2023), designed to reduce dependency on fully annotated datasets. Despite these efforts, such methods often struggle with robustness and generalizability, limiting their effectiveness in practical agricultural applications (Singh et al., 2024). Complexities such as occlusions, irregular fruit structures, and overlapping clusters further hinder segmentation performance, emphasizing the urgent need for comprehensive real-field datasets that better represent the challenges of dense crop analysis.

Recent advancements in Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have significantly improved semantic object detection, offering robust solutions for both general-purpose and agriculture-specific applications. For instance, Abdalla et al. (2019) fine-tuned a CNN-based model for semantic segmentation of oilseed crops under high weed pressure. While their approach yielded promising results, its heavy reliance on transfer learning limited adaptability to datasets with distinct characteristics. Similarly, Peng et al. (2020) employed DeepLabv3+ (DLv3+) to segment litchi tree branches, demonstrating effectiveness in tree structure segmentation. Their model struggled with occlusions and overlapping clusters. In another study, Sun et al. (2021) utilized DLv3+ with shape constraint level sets to detect flowers in apple, peach, and pear crops. Although the use of geometric constraints improved segmentation accuracy, the approach was limited by the diversity of the dataset and its focus on the flowering stage, reducing its applicability to broader agricultural scenarios.

Depth-aware approaches have been explored to enhance segmentation in dense agricultural environments. Wang et al. (2024b) introduced a depth-aware model to detect concealed crops in dense fields, leveraging depth information and deep learning to address the challenge of overlapping vegetation. In a related study, Wang et al. (2024a) refined the DLv3+ algorithm for high-resolution remote sensing images, incorporating custom post-processing steps to enhance segmentation accuracy. Despite these advancements, challenges such as dense object overlap and occlusions remain partially addressed. Maheswari et al. (2022) proposed an intelligent yield estimation framework using the SegNet (SN) model (Badrinarayanan et al., 2017) for tomato crop segmentation, achieving high precision. Similarly, Kamal et al. (2022) implemented a fully convolutional network (FCN)-based approach (Long et al., 2015) for weed and crop segmentation, integrated with IoT-based agricultural monitoring.

Comparative studies have also examined the strengths and limitations of CNN-based architectures. Abd Abd Almisreb et al. (2022) evaluated CNN models for the segmentation of plant leaf disease, providing insights into their performance. However, the study did not explore transformer-based architectures or hybrid models that integrate CNNs with other deep learning paradigms, which could further enhance segmentation accuracy in complex agricultural environments. Transformer-based architectures have also demonstrated potential in precision agriculture. Zhang et al. (2023) employed SegFormer (Xie et al., 2021) to delineate cotton planting areas from high-resolution remote sensing images, highlighting the scalability of transformer-based models. Similarly, Xu et al. (2024) introduced the Multiscale Contextual Swin Transformer (MSC-Swin) for crop image segmentation, effectively capturing multiscale features. Although these transformer-based models have achieved strong results in controlled environments, their generalizability to complex, highly cluttered, and occluded agricultural fields remains largely unexplored.

1.1. Potential research gaps and motivation

Existing agricultural datasets (Shamrat et al., 2024) are predominantly captured in controlled environments, limiting their generalizability to real-field conditions where factors such as variable lighting, occlusions, and structural complexity significantly impact performance. To address this limitation, we introduce a large-scale, real-field dataset collected from Silal's commercial blueberry farm in Al-Foah, Al Ain, UAE, capturing the intricate challenges of dense crop segmentation in practical agricultural settings. Another critical gap exists in the synergy between AI solution providers and experts in the agricultural domain. Many AI-driven solutions lack direct feedback from field experts, resulting in models that do not align with real-world agricultural constraints and operational needs. To mitigate this issue, our work is fully integrated with expert feedback from real-field experts. By incorporating domain expertise and real-world operational insights, we structured our data collection, refined our annotation strategy, and developed a customized detection framework that directly addresses the challenges encountered in practical agricultural applications. Furthermore, based on expert feedback, we expanded our work to develop an LLM-driven recommendation system, customized for real-field needs, further improving decision support for precision agriculture. This approach bridges the gap between theoretical AI advancements and real-world deployment, ensuring seamless integration into large-scale agricultural systems and fostering sustainable AI-powered crop management strategies. Fig. 1 presents a schematic overview of the proposed framework (left) and its performance gains over the baseline models (right). The right panel demonstrates significant improvements across key metrics, underscoring its effectiveness in dense crop segmentation and its potential as a new benchmark for precision agriculture.

1.2. Contribution

This study aims to address the challenges associated with dense crop analysis in wild environments and provides new foundations for the advancement of this research through the following key contributions:

1. Our work bridges the gap between AI solution providers and field experts, integrating direct feedback from real-field experts to collect the dataset and develop a customized AI-driven explainable framework that aligns with practical agricultural needs.
2. We proposed a novel detection framework, SAMConvFormer, which integrates the Segment Anything Model (SAM) with a joint convolutional transformer, addressing key challenges in detecting densely fruited crops in complex agricultural environments. The prompt-driven refinement capabilities of SAM improve boundary delineation, particularly for overlapping clusters, blurred regions, and irregular fruit structures.
3. Extended the detection framework by incorporating a Large Language Model, creating a comprehensive pipeline for yield estimation, ripeness analysis, and actionable crop management strategies, ensuring adaptive decision support for precision agriculture applications.
4. Conducted an in-depth analysis as an ablation study to systematically assess and demonstrate the incremental performance improvements achieved by each component of the proposed framework.
5. The proposed framework will be publicly available to support future research in dense crop segmentation and foster innovation in precision agriculture.

The remainder of this paper is structured as follows: Section 2 details the proposed framework, highlighting its key innovations. Section 3 presents the experimental results with a detailed ablation study, comparing the performance of our model with existing state-of-the-art methods. Section 4 provides an in-depth discussion of the findings, identifying critical gaps and their broader implications. Finally, Section 5 concludes the paper.

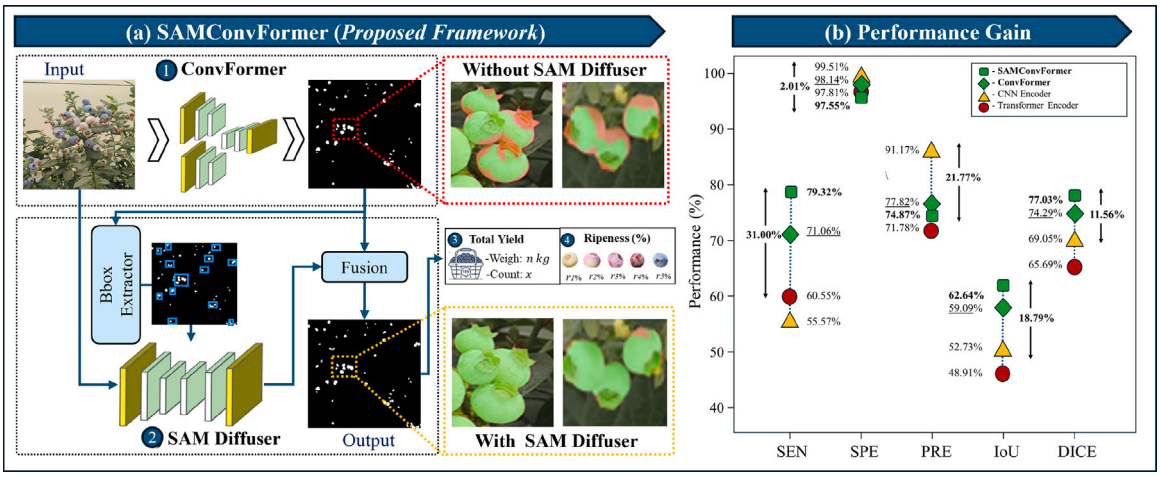


Fig. 1. Schematic overview of the proposed framework (left) and its performance gains over baseline models (right), highlighting its effectiveness in dense crop segmentation and its potential as a benchmark for precision agriculture.

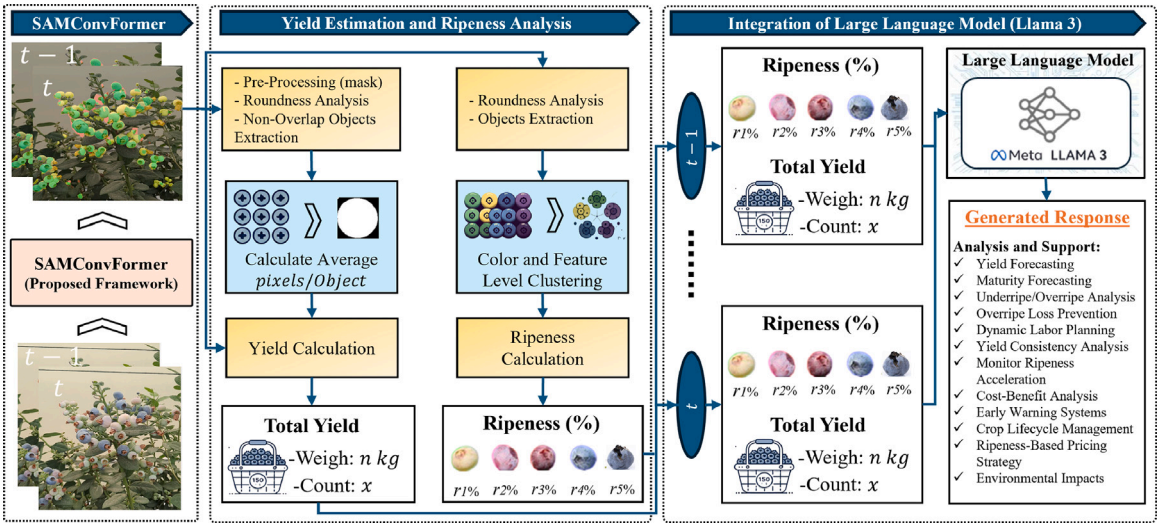


Fig. 2. Workflow of the proposed framework, comprising fruit segmentation with SAMConvFormer, yield estimation and ripeness analysis via a statistical algorithm, and actionable crop management through LLM integration.

2. Proposed framework

The proposed framework comprises three key stages: 1) SAMConvFormer (for fruit segmentation), 2) yield estimation and ripeness analysis, and 3) LLM integration. For fruit segmentation, we introduce the novel SAMConvFormer framework, which integrates the SAM with a joint convolutional transformer to ensure robust crop detection in a real field environment. In the second stage, a statistical analysis-based algorithm is developed to process the segmented masks, enabling accurate yield estimation and ripeness analysis. This stage utilizes segmented results to generate actionable insights into crop yield and maturity. Finally, the integration of LLM extends the pipeline into a decision-support framework for crop management. By leveraging the data generated from yield estimation and ripeness analysis pipeline, the LLM provides actionable recommendations and support for optimized agricultural practices. The overall workflow of the proposed framework is illustrated in Fig. 2, with subsequent subsections detailing each component of the system.

2.1. SamConvFormer

We propose a novel segmentation framework that comprises the following three segments: 1) joint encoder, 2) joint decoder, and 3) SAM diffuser. The joint encoder and decoder (in our first ConvFormer network) integrates the global feature extraction capabilities of ViT (Dosovitskiy, 2020) with the local context modeling strengths of residual CNN (He et al., 2016), enabling precise segmentation in complex and real-world agricultural scenarios. The SAM diffuser further refines the segmentation process by improving boundary delineation, particularly in challenging cases such as overlapping clusters, blurred regions, and irregular fruit structures. The overall architecture and workflow of the proposed SAMConvFormer are illustrated in Fig. 3, with subsequent subsections providing detailed explanations of each component.

2.1.1. Joint encoder

The Joint Encoder integrates feature representations from the transformer encoder, ψ_{ViT_E}, capturing long-range dependencies and global context, with outputs from the residual-based CNN encoder, ψ_{CNN_E}, rich in local spatial details. The features are concatenated along the channel dimension to produce a unified representation, preserving fine-grained spatial information from the CNN encoder while enhancing it with global context from the transformer.

1) Transformer Encoder: The transformer encoder, ψ_{ViT_E}, comprises a total of 12 lightweight transformer encoder blocks (Dosovitskiy, 2020), constructing a compact backbone encoder to extract

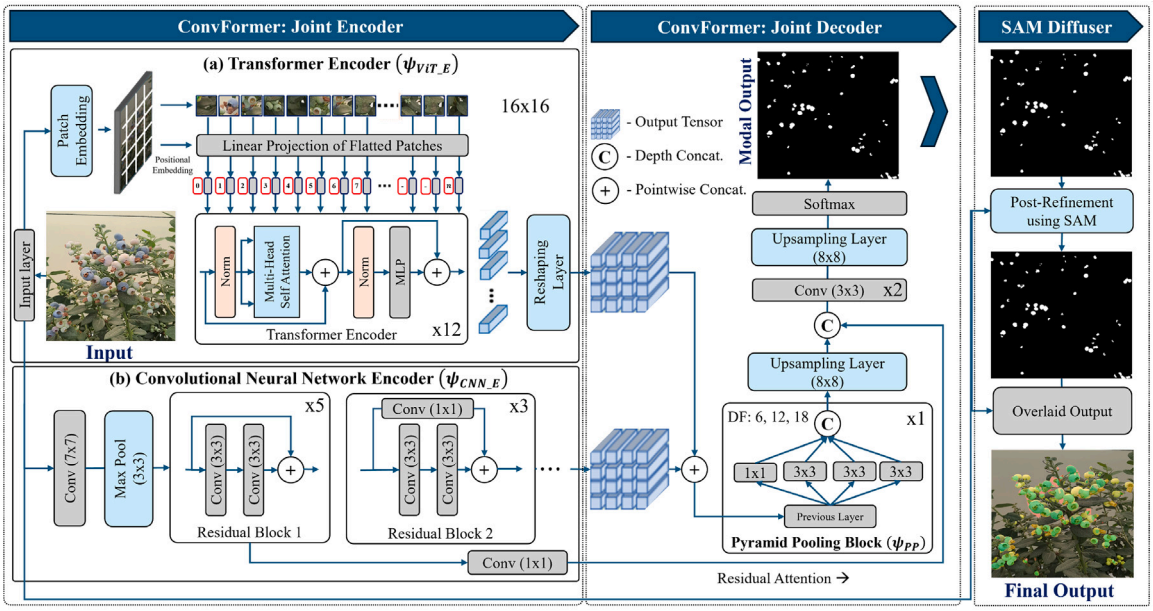


Fig. 3. Architectural overview of the proposed SAMConvFormer framework, which comprises three segments: 1) joint encoder, 2) joint decoder, and 3) SAM diffuser, enabling precise segmentation in complex and real-world agricultural scenarios.

long-range dependencies and global context from the input image $X \in \mathbb{R}^{H\times W\times 3}$. The encoder design mainly addresses key challenges such as occlusions, irregular structures, and clusters that overlap in dense crop segmentation. Initially, the input image X is divided into $k = \frac{H}{s} \times \frac{W}{s}$ non-overlapping patches of size s. Each patch $X_{p_n} \in \mathbb{R}^{s\times s}$ $(n = 1, \ldots, k)$ is projected into embeddings $X_{\text{emb}} \in \mathbb{R}^{k\times d}$ using a linear projection layer as shown in Fig. 3 (transformer encoder ψ_{ViT_E}). Positional embeddings P_{pos} are added to X_{emb} to form the input $X_{\text{in}} = X_{\text{emb}} + P_{\text{pos}}$, preserving spatial relationships. The input embeddings X_{in} are processed through the transformer encoder blocks $N = 12$. Each encoder block comprises a multi-head self-attention mechanism, a feedforward neural network (FFN), residual connections, and layer normalization, ensuring stable training and effective feature extraction. In block l, queries (Q), keys (K), and values (V) are computed from X_{in}^{l-1} as: $Q = X_{in}^{l-1}W_Q$, $K = X_{in}^{l-1}W_K$, $V = X_{in}^{l-1}W_V$, where $W_Q, W_K, W_V \in \mathbb{R}^{d\times d_k}$ are learnable projection matrices, and d_k is the dimension of queries, keys, and values. The scaled dot-product attention and aggregated outputs are computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \tag{1}$$

$$\mathcal{A}_l(X_{in}^{l-1}) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)W_O, \tag{2}$$

where $\text{head}_i = \text{Attention}(Q_i, K_i, V_i)$ and $W_O \in \mathbb{R}^{(h\cdot d_k)\times d}$ is a learnable projection matrix. The aggregated output $\mathcal{A}_l(X_{\text{in}}^{l-1})$ is refined with residual connections and layer normalization:

$$X_{\text{in}}^{l} = \text{LayerNorm}(X_{\text{in}}^{l-1} + \mathcal{A}_l(X_{\text{in}}^{l-1})) + \text{MLP}(X_{\text{in}}^{l-1}), \tag{3}$$

where the MLP includes a single hidden layer with activation. After N blocks, the output features encapsulating long-range dependencies and global context are reshaped into a final tensor $X_{\text{ViT-out}} \in \mathbb{R}^{H'\times W'\times d}$ as: $X_{\text{ViT-out}} = \text{Reshape}(X)$, where $H' = H/s$ and $W' = W/s$ represent the spatial dimensions of the patches. The customized Reshape$(\cdot)$ operation aligns patch-based embeddings into a spatially coherent tensor, preparing the features for further processing. The final features $X_{\text{ViT-out}}$ serve as input to the decoder for dense crop segmentation.

2) Convolutional Neural Network Encoder: The CNN encoder (Fig. 3), inspired by ResNet (He et al., 2016), is designed to address the need to capture fine-grain spatial details, critical for accurate segmentation in complex real-world scenarios. Its architecture leverages residual connections to improve feature learning and mitigate vanishing gradients, ensuring robust generalization to varying conditions. In detail, the CNN encoder, $\psi_{\text{CNN_E}}$, extracts fine-grained local features from the input image $X \in \mathbb{R}^{H\times W\times 3}$ through a series of eight residual blocks, which effectively combine low-level details with high-level features. Each residual block transforms the input X_{in}^{l-1} as:

$$X_{\text{out}}^{l} = \text{ReLU}\big(X_{\text{in}}^{l-1} + \mathcal{F}(X_{\text{in}}^{l-1}, W^l)\big), \tag{4}$$

where $\mathcal{F}(\cdot, W^l)$ represents the residual function parameterized by learnable weights W^l, comprising a series of convolutional, batch normalization, and activation layers. The skip connection X_{in}^{l-1} ensures the preservation of local spatial details while facilitating gradient flow. After residual blocks $N = 8$, the input image $X \in \mathbb{R}^{H\times W\times 3}$ is transformed into the final encoded representation $X_{\text{CNN-out}}$, which encapsulates fine-grained local context, providing an auxiliary input to the decoder for enhanced segmentation under occlusions and cluttered structures.

2.1.2. Joint decoder

The joint decoder is designed to transform the outputs of the transformer encoder ($X_{\text{ViT-out}}$, capturing global context) and the residual encoder ($X_{\text{CNN-out}}$, capturing fine-grained local details) into a high-resolution segmentation mask. To achieve this, the features from both encoders are concatenated along the channel dimension, producing a unified representation X_{joint} as shown in Fig. 3. The concatenation process is mathematically expressed as: $X_{\text{joint}} = \text{Concat}(X_{\text{ViT-out}}, X_{\text{CNN-out}})$, where $X_{\text{ViT-out}} \in \mathbb{R}^{H'\times W'\times d_1}$ and $X_{\text{CNN-out}} \in \mathbb{R}^{H'\times W'\times d_2}$ are the respective outputs of the transformer and CNN encoders, and $X_{\text{joint}} \in \mathbb{R}^{H'\times W'\times(d_1+d_2)}$ represents the combined feature map. This unified representation leverages the strengths of both encoders: the global context provided by the transformer encoder and the fine-grained local details preserved by the CNN encoder. The resulting X_{joint} is processed through a pyramid pooling block, which exploits multiscale features to capture more refined global and local context. It extracts multiscale contextual features from $X_{\text{joint}} \in \mathbb{R}^{H\times W\times d}$ using parallel dilated convolutions with predefined dilation rates. The outputs are concatenated to form the multiscale feature map $X_{PP} \in \mathbb{R}^{H'\times W'\times d}$, expressed as:

$$X_{PP} = \bigcup_{i=1}^{4} \phi_i(X_{\text{joint}}), \tag{5}$$

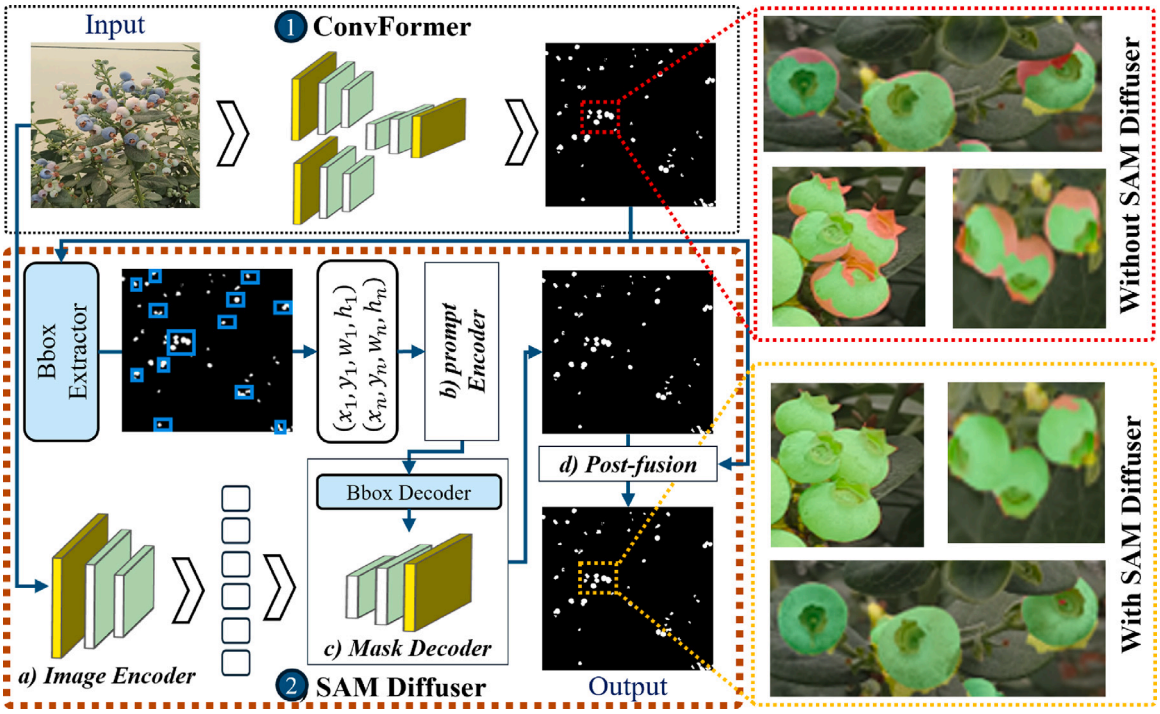


Fig. 4. Architectural overview of the SAM diffuser (with some comparative visual output results) integrated as a secondary refinement stage to improve segmentation accuracy by addressing challenges such as overlapping clusters, blurred regions, and occluded structures.

where $\phi_i(\cdot)$ denotes the ith dilated convolution. The PP block effectively captures fine-grained details and broad global context, addressing challenges in dense crop segmentation. To restore the original image resolution, X_{PP} is progressively upsampled through two transpose convolutional layers: $X_{up} = \text{UpConv}_2(\text{UpConv}_1(X_{PP}))$, where UpConv_1 and UpConv_2 upscale the spatial dimensions by a factor of two. The upsampled feature map X_{up} is then passed through a softmax activation layer and a pixel classification layer to generate the binary segmentation mask, $\hat{M}_{\text{ConvFormer}} = \text{Softmax}(\text{PixelClassify}(X_{up}))$, as the final output of our first stage network (ConvFormer).

2.1.3. SAM diffuser

The SAM model (Kirillov et al., 2023) is integrated into our framework as a secondary refinement stage to enhance segmentation accuracy, particularly in challenging scenarios such as overlapping clusters, blurred regions, and irregular fruit structures. It refines the initial mask $\hat{M}_{\text{ConvFormer}}$ generated by the primary ConvFormer model, addressing cluttered and occluded regions that remain unresolved after the first stage. The SAM diffuser comprises: 1) an image encoder, 2) a prompt encoder that uses input from the bounding box (Bbox) of $\hat{M}_{\text{ConvFormer}}$, 3) a mask decoder, and 4) a post-fusion module as highlighted in Fig. 4 (in red dotted box). The image encoder employs a lightweight transformer (Dosovitskiy, 2020) to process the input image $X \in \mathbb{R}^{H\times W\times C}$, producing an encoded representation X_{en} that encapsulates global and local context, forming a robust foundation for dense segmentation tasks.

Subsequently, the segmentation mask $\hat{M}_{\text{ConvFormer}} \in \mathbb{R}^{H\times W}$ is used in the prompt encoder stage to extract the bounding boxes, parameterized as $\text{Bbox}_i = (x_i, y_i, w_i, h_i)$, where (x_i, y_i) represents the upper left corner and (w_i, h_i) the width and height. The set of all bounding boxes is defined as $\text{Bbox} = \{(x_i, y_i, w_i, h_i)\}_{i=1}^{n}$, with n denoting the number of segmented fruits. These bounding boxes guide the mask decoder to focus on crop regions, mitigating occlusions and clutter by refining false negative regions missed by ConvFormer. The bounding box coordinates are transformed into encoded embeddings $f_{\text{Bbox}} \in \mathbb{R}^{n\times d_{\text{Bbox}}}$ through a bounding box encoder comprising fully connected layers with non-linear activations, where d_{Bbox} is the feature dimensionality. This embedding encapsulates spatial relationships and contextual information, enhancing the precise localization of dense objects.

Finally, the mask decoder refines the encoded image X_{en} (obtained from image encoder in SAM) by integrating bounding box embeddings f_{Bbox}, enhancing segmentation precision in occluded and cluttered regions. X_{en} captures global and local contexts, while f_{Bbox} provides region-specific focus. The fusion of X_{en} and f_{Bbox} is achieved through spatial alignment and projection:

$$F_{\text{fused}} = \text{Concat}(\text{Proj}_X(X_{en}), \text{Proj}_B(f_{\text{Bbox}})), \tag{6}$$

where $\text{Proj}_X(\cdot)$ and $\text{Proj}_B(\cdot)$ project X_{en} and f_{Bbox} into a common feature space. A multi-head attention mechanism is applied to F_{fused} to prioritize relevant features, followed by convolutional layers and upsampling to reconstruct the refined mask $\hat{M}_{\text{SAM Diffuser}}$. The final segmentation mask $M_{\text{SAMConvFormer}}$ is obtained through post-fusion operation: $M_{\text{SAMConvFormer}} = \hat{M}_{\text{ConvFormer}} \cup \hat{M}_{\text{SAM Diffuser}}$. This fusion balances sensitivity and specificity, providing an accurate representation of dense crops and addressing occlusions and field complexities.

2.2. Yield estimation and ripeness analysis

The second stage of the proposed framework employs a statistical analysis-based algorithm (Table 1) that integrates geometric and color-based features to process segmented masks, enabling precise yield estimation and ripeness analysis (as illustrated in Fig. 2). Yield estimation employs a roundness-based analysis to identify non-overlapping objects within the segmented mask, ensuring accurate extraction of individual objects. To improve object delineation, the preprocessing step (as detailed in Table 1) extracts edge information from the original image using Canny edge detection (Luo and Duraiswami, 2008), followed by computing a refined mask that integrates both segmentation and edge maps. Subsequently, morphological operations (Soille and Soille, 2004) are then applied to close small gaps and fill holes, improving object continuity and accuracy. Roundness is quantified using the circularity metric, defined as: $C = \frac{4\pi A}{P^2}$, where C is the circularity,

Table 1
Workflow algorithm for yield estimation and ripeness analysis.

Step	Description and Equations
Yield Estimation Steps:	
1	Extract edge information from the original image using Canny edge detection (Luo and Duraiswami, 2008).
2	Compute refined mask: $M_{\text{refined}} = M_{\text{SAMConvFormer}} \cap M_{\text{edge}}$.
3	Perform morphological operations to close small gaps and fill holes in M_{refined}.
4	Identify objects using roundness analysis: $C = \frac{4\pi A}{P^2}$, exclude objects with $C < 0.8$.
5	Compute A_{avg}: $A_{\text{avg}} = \frac{\sum_{k=1}^{N_{\text{valid}}} A_k}{N_{\text{valid}}}$, where A_k is the area of the kth valid object.
6	Estimate total blueberries: $N = \frac{A_{\text{total}}}{A_{\text{avg}}}$, where $A_{\text{total}} = \sum_{i=1}^{N_{\text{objects}}} A_i$.
7	Compute yield: $Y_{\text{kg}} = N \cdot w_{\text{avg}}$.
8	Average yield across images: $Y_{\text{avg}} = \frac{1}{M} \sum_{j=1}^{M} Y_{\text{kg}}^{(j)}$.
Ripeness Analysis Steps:	
1	Classify objects into ripeness levels r_i $(i = 1, \ldots, 5)$ via clustering.
2	Compute ripeness percentage: $P_{r_i} = \frac{N_{r_i}}{N_{\text{total}}} \cdot 100$.
3	Average ripeness percentage: $P_{r_i,\text{avg}} = \frac{1}{M} \sum_{j=1}^{M} P_{r_i}^{(j)}$.
4	Output Y_{avg} and $P_{r_i,\text{avg}}$.

A is the area of the object and P is the perimeter. Objects with $C < 0.8$ are excluded to focus on well-defined round objects. The average pixel area of valid objects, A_{avg}, is calculated to approximate the area of a single blueberry. The total pixel area of all segmented objects, A_{total}, is used to estimate the total number of blueberries, N, as: $N = \frac{A_{\text{total}}}{A_{\text{avg}}}$. Perimeters are extracted using contour-based methods, such as the Sobel edge detector, and morphological operations are applied to refine object separation. This approach ensures precise yield estimation by focusing on geometric properties, minimizing errors from overlapping or irregular shapes. Using the estimated number of blueberries, N, and the average weight of a single blueberry, w_{avg}, a predefined hyperparameter, the yield in kilograms, Y_{kg}, is computed as: $Y_{\text{kg}} = N \cdot w_{\text{avg}}$. This computation is performed for each input image, and the overall yield is determined by averaging the individual results across the dataset. Unlike methods that infer the size of the fruit from the pixel area (which are sensitive to the distance of the camera), our approach estimates the yield based on the number of objects. We apply geometric filtering (e.g., circularity thresholding) to isolate valid fruit instances and count them directly. This object count is then multiplied by the average weight of a single blueberry, predefined for the specific greenhouse. As such, the accuracy of our method does not depend on maintaining a fixed camera distance, making it suitable for real-field scenarios where image acquisition conditions may vary.

For ripeness analysis, the algorithm considers five ripeness levels: r_1: unripe, r_2: less than half, r_3: half, r_4: ripped, and r_5: over-ripped (as illustrated in Fig. 2). The process involves three steps: 1) geometric features are used to extract segmented objects, 2) color and feature-level clustering classifies objects into ripeness levels, 3) the percentage of objects in each ripeness level is calculated as: $P_{r_i} = \frac{N_{r_i}}{N_{\text{total}}} \cdot 100$, where N_{r_i} is the number of objects in ripeness level r_i, and N_{total} is the total number of objects. The overall ripeness distribution is determined by averaging these percentages across all input images. The algorithm combines results to provide both average yield and ripeness distribution across the dataset. The average yield, Y_{avg}, is computed as: $Y_{\text{avg}} = \frac{1}{M} \sum_{j=1}^{M} Y_{\text{kg}}^{(j)}$, and the average ripeness percentage for each level, $P_{r_i,\text{avg}}$, is calculated as: $P_{r_i,\text{avg}} = \frac{1}{M} \sum_{j=1}^{M} P_{r_i}^{(j)}$, where M is the total number of images, $Y_{\text{kg}}^{(j)}$ is the yield for the jth image, and $P_{r_i}^{(j)}$ is the ripeness percentage for level r_i in the jth image. This approach integrates geometric and color-based features to enable accurate yield estimation and ripeness analysis on real-field data under diverse visual conditions.

2.3. Integration of LLM (llama 3)

This stage extends the proposed framework to a comprehensive pipeline for precision crop management by leveraging the output from the previous stage (yield estimation and ripeness analysis). The integration of LLM transforms quantitative data into actionable recommendations (as listed in Fig. 2) that address key decision-making challenges such as yield forecasting, ripeness management, and labor planning. We derived these recommendations through close collaboration with experienced real-field managers from our industrial partner, Silal—UAE's leading food and technology company. By leveraging domain expertise and real-world operational insights, we ensured that our proposed solution aligns with practical agricultural challenges by bridging the gap between theoretical advances and practical deployment. For this analysis, we used Llama 3 (a lightweight LLM framework) due to its advanced contextual understanding, superior scalability, and ability to process structured prompts effectively, as highlighted in Dubey et al. (2024). Unlike other LLMs, Llama 3 excels in handling domain-specific tasks by leveraging its enhanced fine-tuning capabilities and efficient token utilization, which are critical for datasets with structured agricultural data. Furthermore, its lower computational overhead compared to models such as GPT-4 (Sanderson, 2023) makes it more suitable for practical deployment in agricultural systems, particularly in resource-constrained environments. We fine-tune the LLM for dense crop analysis by initializing it with a domain-specific system prompt. The prompt defines task-specific behavior, guiding the model to analyze blueberry crop data across 12 key aspects (as listed in Fig. 2). By leveraging historical and real-time data, the model generates actionable insights to enhance yield estimation and optimize decision-making in real-world agricultural settings.

During testing, structured prompts are automatically generated based on the outputs of the segmentation and analysis stages. This ensures consistent and interpretable input for the LLM, allowing it to reason over time-series data in a standardized manner. The prompt is constructed using a fixed template, incorporating key variables such as date, estimated total yield in kilograms, and ripeness percentages across five predefined categories ranging from under-ripe to overripe. For a single entry, the general format is defined as: $LLM_{\text{Prompt}} = \left[\text{Date} \quad Y_{\text{avg}} \quad P_{r_1,\text{avg}} \quad \cdots \quad P_{r_5,\text{avg}}\right]$, where Y_{avg} represents the total yield, and $P_{r_i,\text{avg}}$ denotes the average percentage of fruits within ripeness category r_i (for $i = 1, \ldots, 5$). For example, a typical prompt entry is formatted as:

"Date: 2025-06-01 | Yield: 4.32 kg | Ripeness Levels: R1: 15.2%, R2: 23.7%, R3: 35.4%, R4: 18.1%, R5: 7.6%''.

For multiple time entries, prompts are stacked in sequence:

$$LLM_{\text{Prompt}} = \begin{bmatrix} \text{Date}^{(1)} & Y_{\text{avg}}^{(1)} & P_{r_1,\text{avg}}^{(1)} & \cdots & P_{r_5,\text{avg}}^{(1)} \\ \text{Date}^{(2)} & Y_{\text{avg}}^{(2)} & P_{r_1,\text{avg}}^{(2)} & \cdots & P_{r_5,\text{avg}}^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \text{Date}^{(t)} & Y_{\text{avg}}^{(t)} & P_{r_1,\text{avg}}^{(t)} & \cdots & P_{r_5,\text{avg}}^{(t)} \end{bmatrix}, \tag{7}$$

where t is the number of data entries over time. By integrating structured quantitative data with contextual interpretation, LLM effectively transforms the raw output into high-level insights, enabling precise decision-making and optimizing harvesting schedules, labor distribution, and resource allocation.

3. Results and analysis

3.1. Dataset

Existing agricultural datasets are often captured in controlled environments, limiting their applicability to real-world conditions where factors such as variable lighting, occlusions, and structural complexity significantly impact performance. To bridge this gap, all images in the proposed dataset were collected in a commercial indoor greenhouse operated by Silal in Al-Foah, UAE, capturing the real-world complexities of dense crop segmentation (available at figshare). Although the environment is structurally enclosed, it reflects real-field agricultural settings used in large-scale crop production. To introduce realistic variability, data was collected at different locations within the greenhouse and at various times, resulting in significant changes in lighting conditions. These variations include the filtering of natural light through greenhouse panels, the shadows cast by structural elements, and the differences in artificial lighting between zones. This lighting diversity, combined with challenges such as occlusions and background clutter, ensures that the dataset effectively represents real-world agricultural scenarios for robust crop segmentation evaluation. Our dataset comprises four distinct sub-collections, acquired using four different mobile cameras to introduce sensor variability and enhance generalizability. The data collection process was conducted under diverse real-world farm conditions, incorporating challenges such as non-uniform illumination, shadows, occlusions, overlapping clusters, and foliage interference, which are commonly overlooked in controlled datasets.

A total of 520 videos were recorded, capturing the natural variability of real-field conditions and resulting in a total of 141,453 frames. From this extensive collection, we curated 1,195 keyframes, ensuring comprehensive coverage across all recorded videos while eliminating redundancy. Each keyframe was precisely annotated by domain experts, adhering to rigorous standardized protocols to maintain high annotation accuracy, even in challenging scenarios such as occlusions, overlapping clusters, and complex lighting conditions. The final dataset contains pixel-wise segmentation masks for 51,145 individual blueberries, ensuring high annotation precision even in occluded and cluttered regions. Fig. 5 presents representative sample images from each dataset with corresponding ground truth masks overlaid for clarity. By focusing on real-world variability, this dataset provides a rigorous benchmark for evaluating segmentation models in commercial agricultural settings, where conventional data sets – collected in controlled environments – fail to generalize effectively. The scale, diversity, and quality of the annotation contribute to the advancement of segmentation research for high-density crop fields, particularly under challenging agricultural conditions.

3.2. Experimental setup

The proposed model was evaluated in a fully supervised learning framework, using a 70%-10%–20% split for training, validation, and testing, respectively. Fully supervised learning was selected to ensure precise and reliable segmentation on real-field data, establishing a robust foundation for future research and domain-specific fine-tuning while leveraging our pre-trained networks for adaptation to real-world agricultural conditions. We used PyTorch and MATLAB (2024a) with Deep Learning toolbox for the implementation and validation of the proposed and baseline models. The experiments were performed on a high-performance system running Windows 11, with an Intel Core i9 processor, 128 GB RAM, and an NVIDIA RTX A6000 GPU with 48 GB of RAM, providing the computational power required for efficient training and validation.

Model performance was assessed using a comprehensive set of segmentation metrics, including sensitivity (SEN), specificity (SPE), precision (PRE), intersection over union (IoU), and Dice coefficient (DICE), each of which captures different aspects of segmentation quality across scenarios. These metrics collectively provide a comprehensive understanding of the segmentation performance for blueberry detection, capture the accuracy of identifying blueberry and background regions, and quantify the overlap quality between predicted berry masks and ground truth, which is crucial for reliable yield estimation and ripeness analysis. For example, SEN measures the proportion of correctly identified positive pixels: $SEN = \frac{TP}{TP+FN}$, where TP and FN represent true positive and false negative pixel counts, respectively. SPE quantifies the proportion of correctly identified negative (background) pixels: $SPE = \frac{TN}{TN+FP}$, where TN and FP denote true negative and false positive pixel counts. PRE evaluates the correctness of predicted positive pixels: $PRE = \frac{TP}{TP+FP}$, IoU measures the overlap between the predicted and ground truth segmentation: $IoU = \frac{TP}{TP+FP+FN}$, DICE evaluates segmentation similarity and is defined as: $DICE = \frac{2\times TP}{2\times TP+FP+FN}$.

3.3. Model training scheme

The proposed ConvFormer adopts a fully supervised training strategy, while the SAM diffuser operates as a zero-shot model during testing without requiring additional training. To address class imbalance in dense crop segmentation tasks, the ConvFormer uses a weighted cross-entropy loss function (Jadon, 2020) within a hierarchical multi-stage training scheme. Table 2 presents the overall training scheme and workflow of the proposed ConvFormer using hierarchical multi-stage training. This process unfolds in three sequential stages to ensure optimal convergence and robust generalization. In the first stage, the transformer-based segmentation model is trained independently to capture global features. In the second stage, the CNN-based model is trained separately to exploit local spatial features. Finally, the two models are fused to develop our first ConvFormer network, and fine-tuning is performed by freezing 90% of the layers and training only the final layers. This multi-stage strategy outperforms end-to-end training by reducing overfitting and underfitting, particularly with limited data, while improving domain adaptation and generalization, as demonstrated in the ablation study. The training loss at each stage is defined as:

$$\mathcal{L}_{WCE} = \begin{cases} \text{Stage 1: } -\sum_j \left(\alpha M_i[j]\log(\hat{M}_{ViT,i}[j]) + \beta(1-M_i[j])\log(1-\hat{M}_{ViT,i}[j])\right), \\ \text{Stage 2: } -\sum_j \left(\alpha M_i[j]\log(\hat{M}_{CNN,i}[j]) + \beta(1-M_i[j])\log(1-\hat{M}_{CNN,i}[j])\right), \\ \text{Stage 3: } -\sum_j \left(\alpha M_i[j]\log(\hat{M}_{ConvFormer,i}[j]) \right. \\ \qquad \left. + \beta(1-M_i[j])\log(1-\hat{M}_{ConvFormer,i}[j])\right). \end{cases} \tag{8}$$

where $M \in \{0,1\}^{H\times W}$ is the ground-truth mask, $\hat{M} \in [0,1]^{H\times W}$ is the predicted mask, and α, β are weights that balance the contributions of the foreground and background. This hierarchical multi-stage training scheme enables the ConvFormer to adopt the complementary strengths of ViT and CNN, ensuring superior performance and robust generalization in dense crop segmentation tasks.

3.4. Ablation study: Proposed vs baseline

To assess the contributions of different components and configurations in our proposed architecture, we performed a detailed ablation study. These experiments analyze the impact of removing or modifying specific components and training strategies, highlighting their respective roles in achieving high performance. Table 3 explores the effect

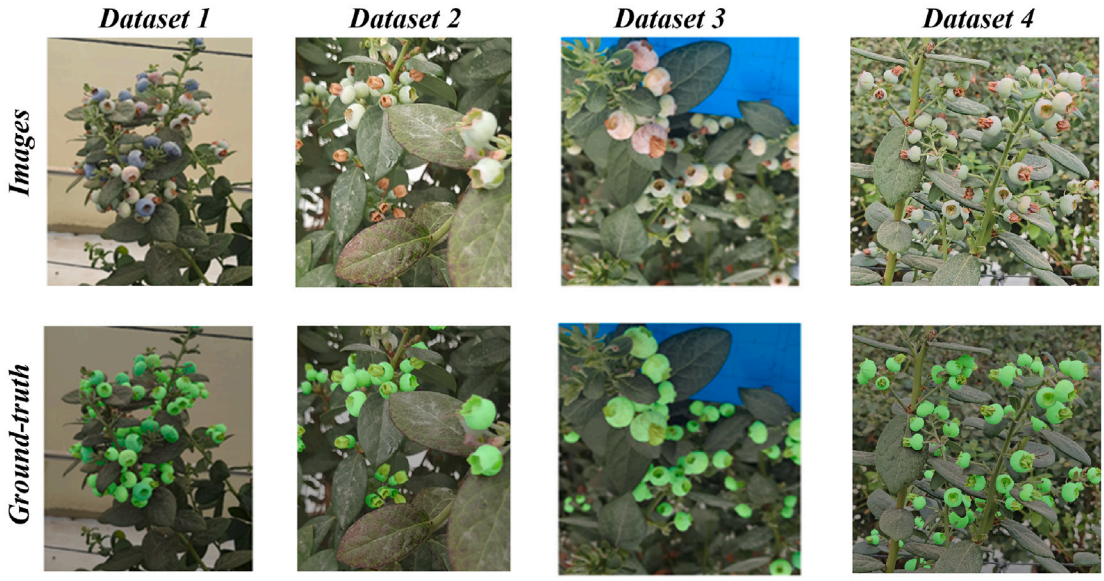


Fig. 5. Sample images from each of the four datasets collected from a commercial blueberry farm, showcasing diverse real-world conditions. Ground truth masks are overlaid to highlight annotations of individual blueberries.

Table 2
Hierarchical multi-stage training scheme of the proposed network.

Step	Description and Equations
Stage 1: Training the ViT-based Model	
1	Load dataset $D = \{(X_i, M_i)\}_{i=1}^{N}$, where $X_i \in \mathbb{R}^{H \times W \times 3}$ are images, and $M_i \in \{0,1\}^{H \times W}$ are ground truth masks.
2	Extract features: $X_{\text{ViT-out}} = \psi_{\text{ViT_E}}(X_i)$, where $X_{\text{ViT}} \in \mathbb{R}^{h \times w \times c}$.
3	Decode features: $\hat{M}_{\text{ViT},i} = \psi_{\text{ViT_D}}(X_{\text{ViT-out}})$.
4	Compute weighted cross-entropy loss: $\mathcal{L}_{\text{ViT}} = -\sum_j \left(\alpha M_i[j] \log(\hat{M}_{\text{ViT},i}[j]) + \beta(1 - M_i[j]) \log(1 - \hat{M}_{\text{ViT},i}[j])\right)$.
5	Update ViT model weights using backpropagation.
Stage 2: Training the CNN-based Model	
1	Load dataset $D = \{(X_i, M_i)\}_{i=1}^{N}$.
2	Extract features: $X_{\text{CNN-out}} = \psi_{\text{CNN_E}}(X_i)$, where $X_{\text{CNN}} \in \mathbb{R}^{h \times w \times c}$.
3	Decode features: $\hat{M}_{\text{CNN},i} = \psi_{\text{CNN_D}}(X_{\text{CNN-out}})$.
4	Compute weighted cross-entropy loss: $\mathcal{L}_{\text{CNN}} = -\sum_j \left(\alpha M_i[j] \log(\hat{M}_{\text{CNN},i}[j]) + \beta(1 - M_i[j]) \log(1 - \hat{M}_{\text{CNN},i}[j])\right)$.
5	Update CNN model weights using backpropagation.
Stage 3: Fine-tuning the Combined Model (ConvFormer)	
1	Integrate ViT and CNN encoders to form the ConvFormer.
2	Extract features: $X_{\text{ConvFormer-out}} = \psi_{\text{ConvFormer_E}}(X_i)$, where $X_{\text{ConvFormer}} \in \mathbb{R}^{h \times w \times c}$.
3	Decode features: $\hat{M}_{\text{ConvFormer},i} = \psi_{\text{ConvFormer_D}}(X_{\text{ConvFormer-out}})$.
4	Fine-tune the network by freezing 90% of the layers and training the last few layers.
5	Compute weighted cross-entropy loss for the combined model: $\mathcal{L}_{\text{ConvFormer}} = -\sum_j \left(\alpha M_i[j] \log(\hat{M}_{\text{ConvFormer},i}[j]) + \beta(1 - M_i[j]) \log(1 - \hat{M}_{\text{ConvFormer},i}[j])\right)$.
6	Update ConvFormer model weights using backpropagation.

of removing the CNN encoder (He et al., 2016), ViT encoder (Dosovitskiy, 2020), and pyramid pooling block from the proposed ConvFormer model (our first stage model). When CNN encoder was excluded, the sensitivity decreased from 71.06% to 60.55%, a relative drop of 14.78%, while the Dice coefficient decreased from 74. 29% to 65. 69% (a relative decrease of 11.58%). These results underscore the critical role of convolutional layers in capturing local spatial information essential for dense crop analysis. The exclusion of the ViT encoder resulted in a sensitivity of 55.57% (a decrease of 21.81% relative to the proposed model), but it achieved the highest specificity (99.51%) and precision (91.17%). These findings suggest that the ViT encoder contributes to enhancing spatial consistency and capturing global context, complementing the convolutional backbone. However, the Dice coefficient decreased to 69. 05% (a relative decrease of 7.05%), indicating an overall decline in performance. Removing the pyramid pooling module led to a sensitivity of 69.20% (a relative drop of 2.62%) and a Dice coefficient of 73.14% (a decrease of 1.55%). This module demonstrated its importance in the aggregation of multi-scale features, as its absence caused a slight performance degradation. To validate the statistical significance of these performance gains, we performed a paired t-test analysis (Kim, 2015) between the proposed ConvFormer and the baseline configurations. The results confirm that the observed improvements are statistically significant, achieving a confidence level of 99% with $p = 0.01$ when CNN encoder was removed and 90% with $p = 0.092$ when the pyramid pooling module was removed. This analysis underscores the robustness and reliability of the proposed framework in addressing the inherent challenges of dense crop segmentation.

The qualitative results in Fig. 6 provide visual confirmation of the contributions of individual components to the proposed architecture. True positive regions, indicated in green, highlight the ability of model to accurately segment the target crop areas. The last column, which represents the results of the proposed ConvFormer model, demonstrates the most consistent segmentation performance, with minimal false negatives in red and false positives in yellow. In contrast, removing

Table 3
Quantitative ablation study evaluating the impact of removing individual components (CNN encoder, ViT encoder, and pyramid pooling block) from the proposed ConvFormer model (our first stage model). **Bold: Best results**, Underline: Second best.

Model	SEN (%)	SPE (%)	PRE (%)	IoU (%)	DICE (%)
$\psi_{\text{ConvFormer}} - \psi_{\text{CNN_E}}$	60.55	97.81	71.78	48.91	65.69
$\psi_{\text{ConvFormer}} - \psi_{\text{ViT_E}}$	55.57	**99.51**	**91.17**	52.73	69.05
$\psi_{\text{ConvFormer}} - \psi_{\text{PP}}$	69.20	98.16	77.57	57.66	73.14
$\psi_{\text{ConvFormer}}$	**71.06**	98.14	77.82	**59.09**	**74.29**

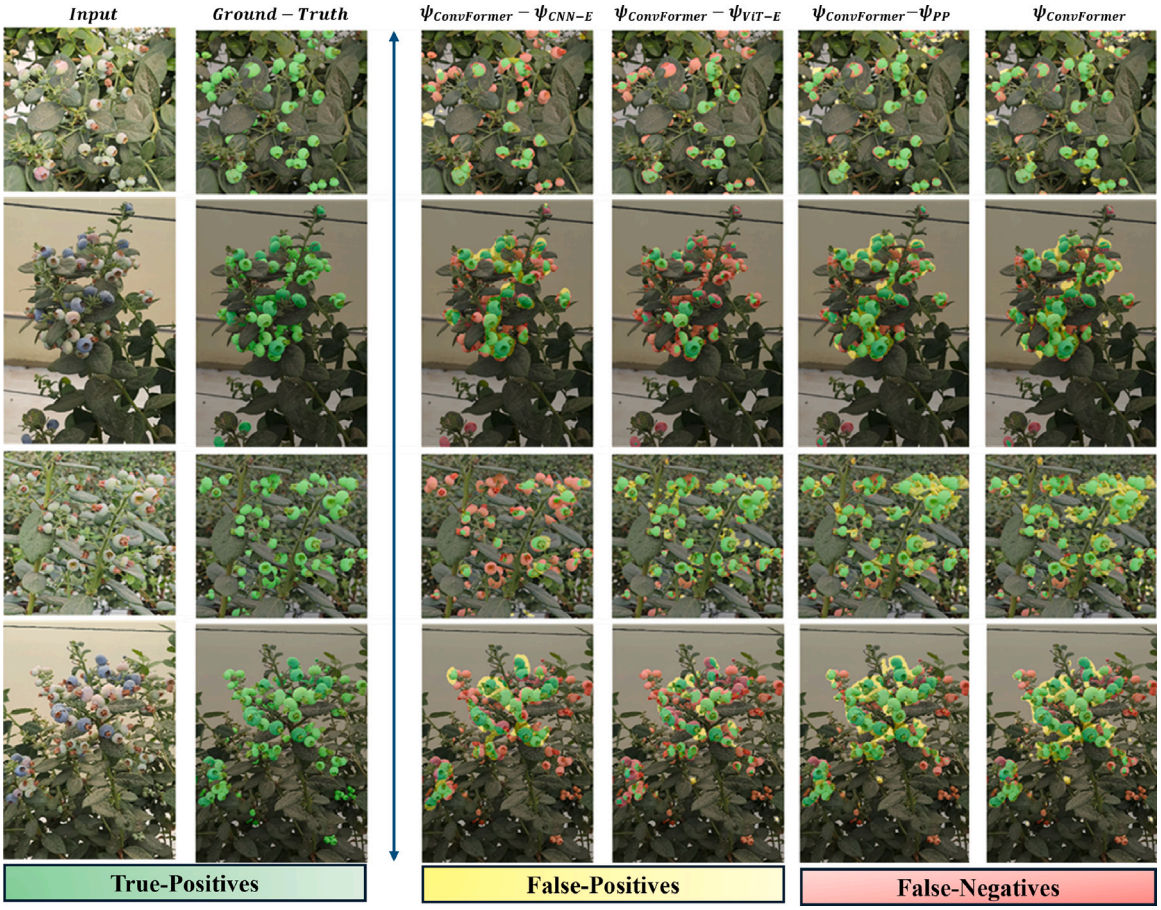


Fig. 6. Visual comparison of segmentation results illustrating the effect of removing individual components (CNN encoder, ViT encoder, and pyramid pooling block) from the proposed ConvFormer model (our first stage model).

the CNN encoder results in more fragmented predictions, leading to an increase in false negatives, particularly in densely packed regions. Similarly, the exclusion of the ViT encoder affects the ability to maintain spatial consistency, resulting in more false positives and gaps in target coverage. The absence of the pyramid pooling module yields slight performance degradations with both false positives and false negatives. These visual observations align with the quantitative results Table 3, reaffirming the critical roles of the CNN encoder, ViT encoder, and pyramid pooling block in achieving robust segmentation performance of our first stage model.

To evaluate the performance and computational efficiency of various CNN-based segmentation models, we conducted a comparative analysis. The results are summarized in Table 4. Among the models, SegNet ($\psi_{\text{SN (VGG16)}}$ and $\psi_{\text{SN (VGG19)}}$) (Badrinarayanan et al., 2017) exhibits the highest computational demands, with parameter counts of 138M and 144M, respectively. Despite its complexity, $\psi_{\text{SN (VGG16)}}$ demonstrates the lowest sensitivity (36. 44%) and the Dice coefficient (49. 28%), indicating suboptimal segmentation performance. $\psi_{\text{SN (VGG19)}}$ offers only a marginal improvement in sensitivity (42.85%), suggesting diminishing returns with deeper architectures. DeepLabv3+ with MobileNet ($\psi_{\text{DLv3+ (MN)}}$) (Chen et al., 2018; Sandler et al., 2018), with only 3.2M parameters, achieves high specificity (99.58%) and precision (91.54%), but its sensitivity (49.87%) and Dice coefficient (64.56%) remain lower than those of larger models. A more complex alternative, DeepLabv3+ with Inception-ResNet ($\psi_{\text{DLv3+ (IRN)}}$) (Chen et al., 2018; Szegedy et al., 2017), enhances the sensitivity (50. 98%) and the Dice coefficient (65. 50%) at the cost of increased computational load (54M parameters). FCN-32S ($\psi_{\text{FCN (32S)}}$) (Long et al., 2015) strikes a balance between complexity and segmentation quality, utilizing 134M parameters to achieve the second-best Dice coefficient (66.37%) and IoU (49.66%). The highest performance is observed with DeepLabv3+ paired with ResNet ($\psi_{\text{DLv3+ (RN)}}$) (Chen et al., 2018; He et al., 2016), which, with 44M parameters, achieves the highest Dice coefficient (69.05%) and IoU (52.73%), demonstrating a favorable trade-off between computational efficiency and segmentation accuracy.

Table 4
Performance comparison of various CNN-based segmentation models to identify the most suitable convolutional encoder for the proposed framework. **Bold: Best results**, Underline: Second best.

Model	SEN (%)	SPE (%)	PRE (%)	IoU (%)	DICE (%)
$\psi_{SN\ (VGG16)}$ (Badrinarayanan et al., 2017)	36.44	98.95	76.11	32.70	49.28
$\psi_{SN\ (VGG19)}$ (Badrinarayanan et al., 2017)	42.85	98.89	77.99	38.22	55.31
$\psi_{DLv3+\ (MN)}$ (Chen et al., 2018; Sandler et al., 2018)	49.87	**99.58**	91.54	47.67	64.56
$\psi_{DLv3+\ (IRN)}$ (Chen et al., 2018; Szegedy et al., 2017)	50.98	99.57	**91.57**	48.69	65.50
$\psi_{FCN\ (32S)}$ (Long et al., 2015)	54.73	99.06	84.29	49.66	66.37
$\psi_{DLv3+\ (RN)}$ (Chen et al., 2018; He et al., 2016)	**55.57**	99.51	91.17	**52.73**	**69.05**

Table 5
Performance and computational trade-offs of ViT-based segmentation encoders. **Bold: Best results**.

Model	SEN (%)	SPE (%)	PRE (%)	IoU (%)	DICE (%)
$\psi_{ViT\ (Tiny)}$ (Xie et al., 2021)	**60.55**	97.81	71.78	48.91	65.69
$\psi_{ViT\ (Small)}$ (Xie et al., 2021)	60.19	**98.15**	**74.91**	**50.09**	**66.75**

Therefore, we adopted $\psi_{DLv3+\ (RN)}$ a convolutional backbone to develop our model.

To further assess the qualitative performance of these models, we present visual segmentation results in Fig. 7. The figure illustrates how different CNN backbones handle segmentation across various test samples. True positive regions, false positives, and false negatives are highlighted to highlight the strengths and limitations of each model. SegNet-based models ($\psi_{SN\ (VGG16)}$ and $\psi_{SN\ (VGG19)}$) (Badrinarayanan et al., 2017) struggle to accurately delineate target regions, leading to extensive false negatives. DeepLabv3+ with MobileNet ($\psi_{DLv3+\ (MN)}$) (Chen et al., 2018; Sandler et al., 2018) exhibits efficiency in reducing false positives, but does not capture all relevant regions due to its limited sensitivity. DeepLabv3+ with Inception-ResNet ($\psi_{DLv3+\ (IRN)}$) (Chen et al., 2018; Szegedy et al., 2017) achieves a better true positive coverage, but introduces false positives in densely packed areas. The FCN-32S model ($\psi_{FCN\ (32S)}$) (Long et al., 2015) provides a more balanced segmentation, capturing finer details while mitigating false negatives and false positives. Finally, DeepLabv3+ with ResNet ($\psi_{DLv3+\ (RN)}$) (Chen et al., 2018; He et al., 2016) delivers the most robust qualitative results, effectively segmenting sparse and dense regions with minimal errors. These observations further validate our choice of $\psi_{DLv3+\ (RN)}$ as the optimal convolutional backbone for our proposed model.

Table 5 presents a comparative analysis of the performance and computational trade-offs between different ViT-based encoders (Xie et al., 2021), focusing on their size variants (tiny vs. small). The tiny ViT model, with approximately 5.7 million parameters, achieves a Dice coefficient of 65.69% and an IoU of 48.91%, demonstrating competitive performance despite its compact architecture. Scaling up to the small ViT variant, which contains approximately 22.1 million parameters (287.72% more than the tiny ViT model), yields only marginal improvements, with a Dice coefficient of 66.75% and an IoU of 50.09%. The relative performance gain remains modest, with only a 1.61% increase in the Dice coefficient and a 2.41% improvement in IoU. Given that the tiny ViT model achieves comparable accuracy with significantly lower computational complexity, it presents a more efficient choice, effectively balancing segmentation performance and resource constraints. Therefore, we leverage the advantages of this lightweight ViT encoder as the backbone for our proposed model.

Finally, Table 6 compares the performance of end-to-end training and hierarchical multi-stage training, as described in Section 3.3. Although end-to-end training achieves high precision (81.77%) and specificity (98.65%), it exhibits lower sensitivity (65. 84%), Dice coefficient (72. 94%) and IoU (57. 41%) compared to the hierarchical multistage approach. In contrast, hierarchical multi-stage training yields notable improvements, with a Dice coefficient of 74.29% (+1.85%), IoU of 59.09% (+2.68%), and sensitivity of 71.06% (+7.91%). These results demonstrate the effectiveness of progressive fine-tuning in refining segmentation performance, validating the advantages of a structured, multi-stage optimization strategy over conventional end-to-end training.

In conclusion, the ablation study highlights the critical role of integrating CNN, ViT, and PP modules, along with the effectiveness of hierarchical multi-stage training. These components collectively improve segmentation accuracy, enabling the proposed model to achieve state-of-the-art performance in dense crop analysis.

3.5. Ablation study: Quantitative analysis of SAM diffuser

To evaluate the robustness of SAMConvFormer under challenging conditions such as variable lighting, occlusion, and object overlap, present in our custom dataset, we conducted a detailed ablation study. The SAM diffuser module is central to addressing these challenges by refining segmentation masks based on adaptive prompt-aware feature fusion. It leverages bounding-box guidance from ConvFormer to enhance spatial localization and mask refinement, especially in cluttered or ambiguous regions where traditional methods tend to fail. The quantitative results in Tables 7 highlight the impact of integrating the SAM diffuser into ConvFormer and the effect of the variations in the bounding box scale on the SAMConvFormer segmentation performance. The boundary boxes generated by ConvFormer output, along with the input images, were passed to the SAM diffuser module with scale factors of 0.8 (−20%), 0.9 (−10%), 1.0 (default), 1.1 (+ 10%), and 1.2 (+ 20%) to assess the segmentation precision. Scaling influences segmentation by compensating for bounding box prediction errors. Smaller boxes increase false negatives, while larger ones introduce false positives. Sensitivity peaks at 81.66% at 1.2 (+20%), a 14.95% gain over ConvFormer (71.06%), while specificity is highest at 97. 83% with a scale of 0.8 (- 20%), confirming an inverse relationship between the two metrics. SAMConvFormer achieves its best segmentation performance on the default scale (1.0), with IoU at 62.64% (+5.99%) and Dice at 77.03% (+3.69%). A slight gain occurs at 1.1 (+ 10%), where the IoU reaches 62. 00% and Dice 76. 55%, suggesting improved coverage without excessive over-segmentation. The precision is highest with ConvFormer alone (77.82%) and decreases with the SAM diffuser, reaching 75. 88% at 0.8 (- 20%) and 69. 70% at 1.2 (+ 20%) due to oversegmentation. These findings confirm that SAMConvFormer (our final proposed model) performs optimally at the default scale (1.0), with minor benefits at 1.1 (+10%), but experiences reduced precision at higher scaling levels.

Fig. 8 compares the segmentation output of ConvFormer (first-stage model) and SAMConvFormer (final model), with zoomed-in views highlighting performance on individual fruits. SAMConvFormer (rightmost column) achieves superior segmentation accuracy, particularly in reducing false negatives. While ConvFormer struggles with missed detections in clustered or occluded regions, SAMConvFormer effectively



Fig. 7. Qualitative comparison of segmentation performance across different CNN-based models. The results validate the choice of $\psi_{\text{DLv3+ (RN)}}$ (Chen et al., 2018; He et al., 2016) as the optimal convolutional encoder for our proposed framework.

Table 6
Comparison of end-to-end training and hierarchical multi-stage training, demonstrating the advantages of progressive fine-tuning. **Bold: Best results**.

Training scheme	SEN (%)	SPE (%)	PRE (%)	IoU (%)	DICE (%)
End-to-End (Gordo et al., 2017)	65.84	**98.65**	**81.77**	57.41	72.94
Hierarchical Multi-Stage	**71.06**	98.14	77.82	**59.09**	**74.29**

Table 7
Impact of bounding box scaling on the segmentation performance of SAMConvFormer. The table compares different scale factors applied to bounding boxes generated by ConvFormer and their effect on sensitivity, specificity, precision, IoU, and Dice coefficient. **Bold: Best results**, Underline: Second best.

Model	Scale factor	SEN (%)	SPE (%)	PRE (%)	IoU (%)	DICE (%)
$\psi_{\text{ConvFormer}}$	–	71.06	**98.14**	**77.82**	59.09	74.29
$\psi_{\text{SAMConvFormer}}$	0.8	74.40	97.83	75.88	60.17	75.13
$\psi_{\text{SAMConvFormer}}$	0.9	76.64	97.75	75.80	61.57	76.22
$\psi_{\text{SAMConvFormer}}$	1.0	79.32	97.55	74.87	**62.64**	**77.03**
$\psi_{\text{SAMConvFormer}}$	1.1	80.62	97.24	72.86	62.00	76.55
$\psi_{\text{SAMConvFormer}}$	1.2	**81.66**	96.74	69.70	60.27	75.21

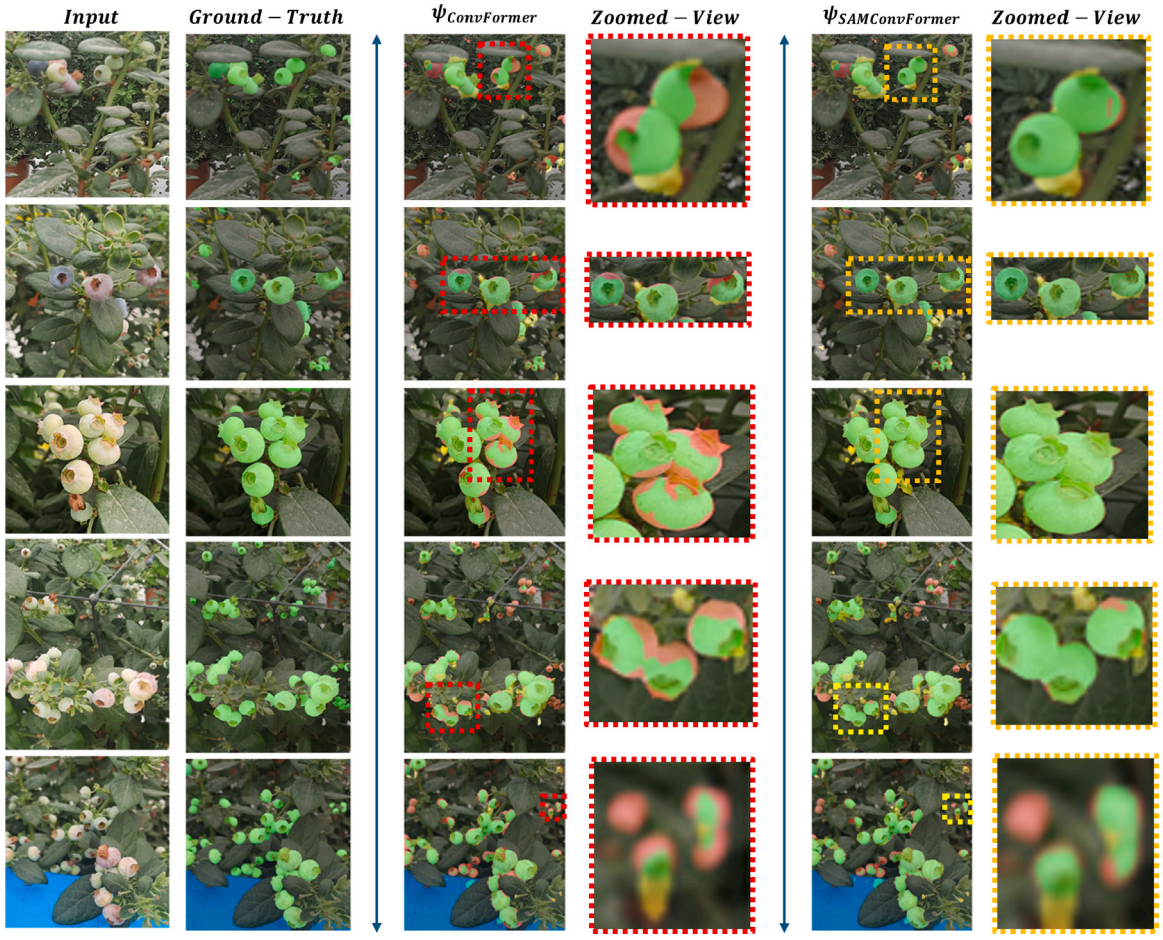


Fig. 8. Qualitative comparison of segmentation outputs from ConvFormer (first-stage model) and SAMConvFormer (final model).

captures more fruit instances, improving sensitivity. The results visually confirm the enhancements introduced by the SAM diffuser module to refine object boundaries and minimize segmentation errors.

Table 8 compares the performance of common post-refinement techniques applied to our first-stage model, evaluating ConvFormer with morphological refinement ($\psi_{\text{ConvFormer}}+\psi_{\text{Morphological}}$) (Abeyrathna et al., 2022; Salvi et al., 2021) against ConvFormer with the SAM Diffuser module ($\psi_{\text{ConvFormer}}+\psi_{\text{SAM Diffuser}}$). The results indicate that the integration of the SAM diffuser significantly enhances the segmentation accuracy in all key metrics. Our proposed ConvFormer model with SAM Diffuser achieves the highest sensitivity (79.32%), outperforming morphological refinement (74.19%), demonstrating improved detection of fruit regions and reduced false negatives. IoU (62.64%) and Dice coefficient (77.03%) also surpass those of morphological refinement (59.24% IoU, 74.40% Dice), indicating more accurate boundary refinement and segmentation consistency. The precision remains comparable, with the SAM Diffuser approach achieving 74.87% versus 74.61% for morphological refinement, suggesting similar false positive rates. The specificity is slightly higher for the morphological refinement (97.68%) compared to the SAM diffuser method (97.55%), indicating a better exclusion of non-fruit regions at the cost of increased false negatives. Overall, the results confirm that the integration of the SAM Diffuser module provides a more balanced and effective post-refinement strategy, significantly improving sensitivity, IoU, and Dice scores while maintaining competitive precision and specificity.

3.6. Ablation study: Yield estimation and ripeness analysis

To assess the effectiveness of the proposed framework in yield estimation and ripeness analysis, a comprehensive ablation study was conducted, comparing predicted values with those derived from ground-truth segmentation masks using the same computational approach. For yield estimation, the predicted average yield $Y_{\text{avg}} = 25.67$ kg was validated against ground-truth yield values (23.37 kg) computed from manually annotated segmentation masks using the same object-counting approach. This comparison yielded an average precision of 90.14%, confirming the reliability of the proposed framework in estimating real-field crop yield. This high accuracy demonstrates the ability of the framework to accurately estimate crop yield, with minimal errors in fruit detection and segmentation. The quantitative results presented in Fig. 9 illustrate the distribution of the average ripeness percentage $P_{r_i,\text{avg}}$ across five maturity levels (r_1 to r_5). The analysis compares the predicted ripeness levels with actual ripeness values obtained from ground-truth segmentation masks using the same ripeness estimation algorithm. The proposed framework demonstrates high precision in ripeness classification at all levels. The predicted ripeness distribution aligns closely with the ground-truth segmentation results, confirming the model's ability to effectively capture variations in fruit maturity. Specifically, the framework achieved an average percentage accuracy of ripeness of 98. 24% for unripe fruits ($P_{r_1,\text{avg}}$), 95.68% for less than half ripe ($P_{r_2,\text{avg}}$), 97.74% for half-ripe ($P_{r_3,\text{avg}}$), 99.34% for fully ripe

Table 8
Comparison of post-refinement models, evaluating ConvFormer with morphological processing (commonly used) and the SAM Diffuser module (ours). The results demonstrate that integrating the SAM Diffuser significantly improves sensitivity, IoU, and Dice coefficient while maintaining competitive precision and specificity. **Bold: Best results**.

Model	SEN (%)	SPE (%)	PRE (%)	IoU (%)	DICE (%)
$\psi_{\text{ConvFormer}} + \psi_{\text{Morphological}}$ (Abeyrathna et al., 2022; Salvi et al., 2021)	74.19	**97.68**	74.61	59.24	74.40
$\psi_{\text{ConvFormer}} + \psi_{\text{SAM Diffuser}}$	**79.32**	97.55	**74.87**	**62.64**	**77.03**

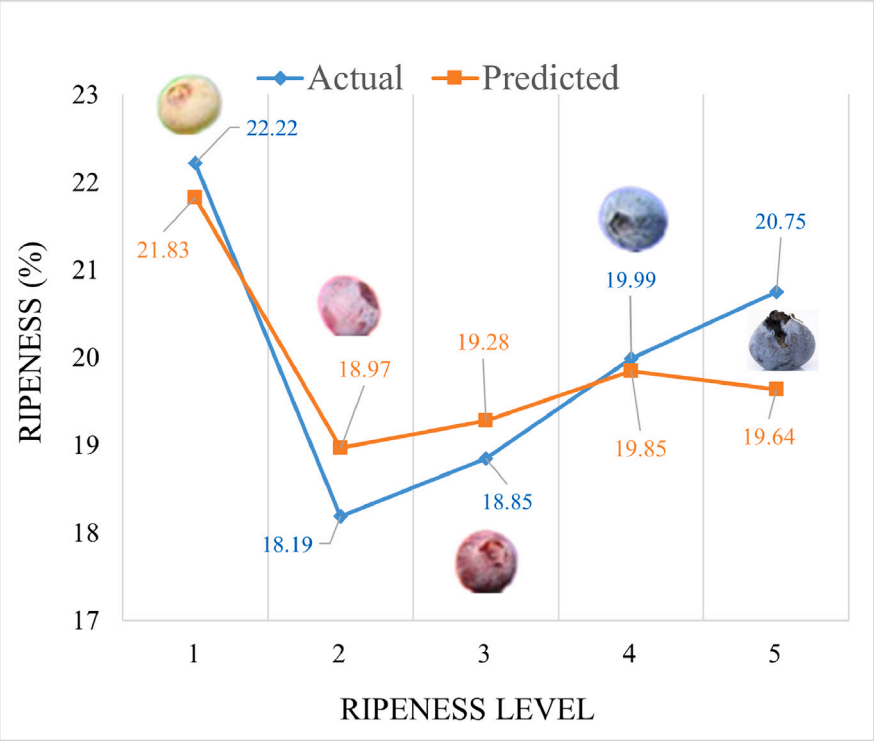


Fig. 9. Illustration of average ripeness distribution $P_{r_i,\text{avg}}$ across five maturity levels (r_1 to r_5), comparing predicted ripeness with ground-truth segmentation results using the same estimation algorithm.

($P_{r_4,\text{avg}}$), and 94.66% for overripe fruits ($P_{r_5,\text{avg}}$). The overall average precision at all ripeness levels is 97. 13%, indicating strong generalization of the model and reliable estimation of maturity in various fruit conditions. These results highlight the robustness of the proposed framework in ripeness estimation, demonstrating its high precision in identifying subtle color transitions and maturity variations.

3.7. Comparison

This section compared the performance of our proposed model (SAMConvFormer) with five state-of-the-art segmentation methods widely used in various general as well as agricultural segmentation tasks (Shamrat et al., 2024; Xu et al., 2024; Sun et al., 2021; Zhang et al., 2023; Peng et al., 2020). Table 9 presents the performance comparison of various methods alongside our proposed SAMConvFormer. Our model outperforms all baselines, achieving the highest Dice Coefficient of 77.03% and IoU of 62.64%. This marks a notable improvement of 11.97% in Dice Coefficient and 17.62% in IoU over the second-best approach (Peng et al., 2020), which achieved 68.08% Dice Coefficient and 51.6% IoU. The SAMConvFormer sensitivity was also ranked as the best performing, showing a significant improvement of 9.64% relative to the results obtained by Shamrat et al. (2024) with a sensitivity of 71.67%. This improvement highlights the superior capability of our proposed model in detecting true positives by handling occlusions, clutter, and complex spatial structures. To ensure that this performance boost was statistically significant, we performed a paired t-test comparing the SAMConvFormer model and the second-best method (Peng et al., 2020). The results demonstrated a significant improvement with a confidence level of 99% ($p < 0.01$), showing that our model is robust and reliable in tackling the challenges of dense crop segmentation. Compared to the third-best model (Zhang et al., 2023) with Dice coefficient of 66.75% and IoU of 50.09%, the proposed model achieved relative improvements of 13.35% in Dice coefficient and 20.04% in IoU. Similarly, sensitivity improved by 24.12% compared to SegFormer (Xie et al., 2021). Although the proposed model excels in improving sensitivity and overall segmentation performance, there was a slight reduction in specificity and precision, this is because the model is focused on minimizing false negatives. The proposed model has achieved significant advances in dense crop segmentation with the highest sensitivity, IoU, and Dice Coefficient values compared to the state-of-the-art methods. The addition of the SAM Diffuser module effectively improves segmentation performance, particularly in occluded and cluttered regions, without sacrificing a good balance in the trade-off between sensitivity and specificity.

The qualitative results in Fig. 10 complement the quantitative findings, comparing segmentation outputs from state-of-the-art models with the proposed SAMConvFormer. Unet (Ronneberger et al., 2015) detects fruit regions but suffers from excessive false negatives, often missing small or occluded fruits. MSC-Swin Xu et al. (2024) improves boundary detection but introduces false positives, misclassifying background

Table 9
Quantitative comparison of the proposed SAMConvFormer model with state-of-the-art methods for crop segmentation. The results highlight the significant performance gains of the proposed model across various metrics. **Bold: Best results**, Underline: Second best.

Author	Model	SEN (%)	SPE (%)	PRE (%)	IoU (%)	DICE (%)
Shamrat et al. Shamrat et al. (2024)	Unet (Ronneberger et al., 2015)	71.67	92.10	45.47	38.54	55.64
Xu et al. Xu et al. (2024)	MSC-Swin Xu et al. (2024)	49.25	98.57	76.02	42.63	59.78
Sun et al. Sun et al. (2021)	DLv3+ (ResNet50) (Chen et al., 2018; He et al., 2016)	46.77	**99.44**	**88.44**	44.08	61.19
Zhang et al. Zhang et al. (2023)	SegFormer (Xie et al., 2021)	60.19	98.15	74.91	50.09	66.75
Peng et al. Peng et al. (2020)	DLv3+ (Xception) (Chen et al., 2018; Chollet, 2017)	56.31	99.16	86.06	51.60	68.08
Proposed	**SAMConvFormer**	**79.32**	97.55	74.87	**62.64**	**77.03**

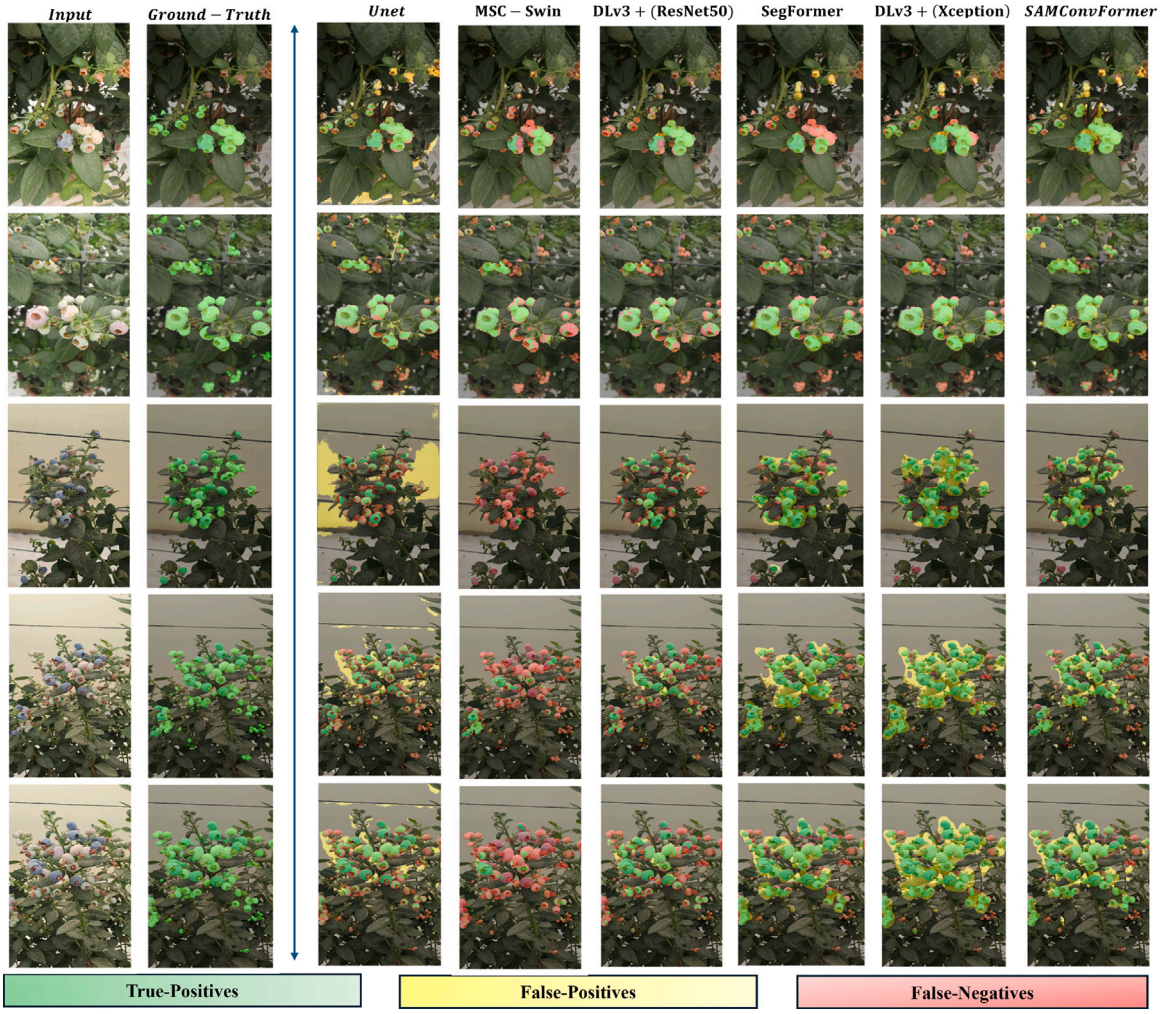


Fig. 10. Qualitative comparison of the proposed SAMConvFormer model with state-of-the-art methods for crop segmentation. The first column shows input images and corresponding ground-truth overlays, while subsequent columns display segmentation results.

elements as fruits. DLv3+ (ResNet50) (Chen et al., 2018; He et al., 2016) produces sharp segmentations but over-segments regions with high texture similarity, increasing false positives. SegFormer (Xie et al., 2021) localizes fruits well but struggles with clustered instances, leading to merging errors and false negatives. DLv3+ (Xception) (Chen et al., 2018; Chollet, 2017) balances segmentation accuracy but still misses some dense fruit regions. Finally, the proposed SAMConvFormer (rightmost column) outperforms all models, maintaining precise fruit boundaries, minimizing false positives, and effectively detecting occluded or clustered fruits with minimal false negatives. These results reinforce its superior quantitative performance, demonstrating its robustness for accurate fruit segmentation in challenging agricultural environments.

4. Discussion

4.1. Key findings

This study takes advantage of recent advances in AI to enhance contextual understanding and decision making in dense crop analysis. We propose SAMConvFormer+LLM, a comprehensive framework that integrates SAM, a Joint Convolutional Transformer, and LLMs to improve dense crop segmentation and AI-driven decision-making in precision agriculture. By combining CNNs for local feature extraction, ViTs for global context modeling, and SAM for segmentation refinement, the framework achieves precise boundary delineation, particularly in overlapping clusters and occluded regions, addressing key

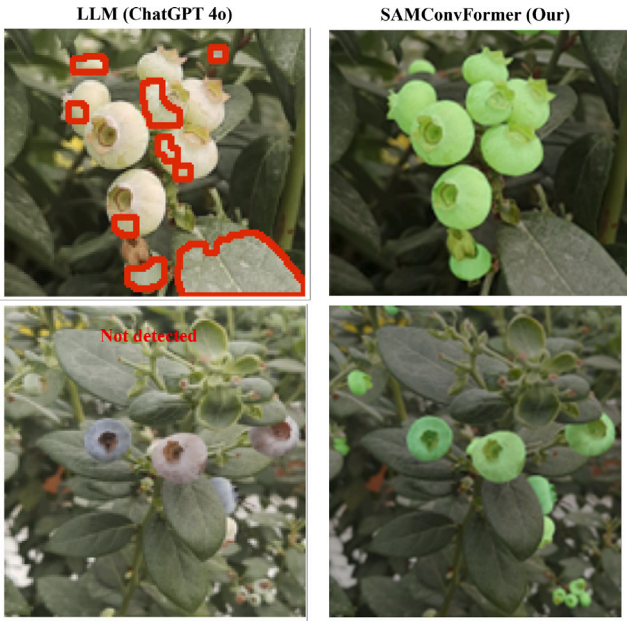


Fig. 11. Comparison between standalone LLM output (ChatGPT-4o, left panel) and the proposed SAMConvFormer+LLM pipeline (right panel). The integrated approach leverages accurate segmentation results to enable precise yield and ripeness analysis, whereas the LLM alone fails to localize fruits reliably.

challenges in real-world agricultural environments. The integration of LLM with post-processing techniques significantly enhances automation in yield estimation and ripeness analysis, enabling structured crop monitoring and informed decision-making. Unlike standalone LLMs, which struggle to accurately detect fruits from raw images due to the absence of spatial and instance-level understanding, our framework first leverages precise segmentation from the SAMConvFormer model to extract object-level information. This structured data is then passed to the LLM for downstream analysis. Fig. 11 highlights this distinction, comparing the performance of a standalone LLM (ChatGPT-4o, left panel) with our proposed integrated approach (right panel). LLM alone cannot accurately detect the boundaries of the fruits, which is essential for reliable estimation of yield and ripeness. In contrast, the proposed pipeline provides the LLM with pre-segmented, high-quality object data, enabling accurate, context-aware recommendations.

Beyond the estimation of yield and ripeness, the LLM component also generates actionable insights for labor planning, harvesting strategy, and sustainable resource allocation, making the pipeline more adaptable and practical for precision agriculture in real field. Fig. 12 presents the results of our LLM-driven framework, demonstrating its effectiveness in yield estimation, ripeness analysis, and decision support. By providing structured insights, the framework significantly enhances data-driven decision-making, offering farmers detailed analytical support for optimizing harvest strategies. To validate the practical significance of LLM-generated recommendations, we conducted a rigorous evaluation in collaboration with experienced real-field managers from our industrial partner, Silal—UAE. As shown in Fig. 12, key insights highlighted in green represent meaningful information validated by industry experts. This expert feedback underscores the relevance and reliability of the proposed framework in addressing real-world agricultural challenges. Using domain expertise and real-world operational insights, we ensured that the framework aligns with practical field conditions. This synergy between AI-driven analysis and expert agricultural knowledge facilitates the development of a robust solution that improves resource management, yield estimation accuracy, and strategic decision-making.

4.2. Limitations

Although the proposed SAMConvFormer+LLM framework achieves state-of-the-art performance in dense crop segmentation and AI-driven decision-making, certain limitations need further exploration to enhance its robustness and scalability. One key limitation is the slight reduction in specificity due to the inclusion of the SAM refinement network, which prioritizes sensitivity to improve boundary delineation. While the post-fusion process mitigates this trade-off to some extent, a more balanced refinement strategy may further enhance segmentation accuracy. Future work could explore adaptive weighting mechanisms or uncertainty-aware fusion techniques to dynamically balance sensitivity and specificity based on contextual information.

Another challenge lies in the computational complexity introduced by the multi-stage refinement process. The proposed model consists of 26.3 million trainable parameters, with a memory footprint of 78.6 MB. The average inference time per image is 0.185 s without the SAM module and 0.982 s with SAM included. The integration of CNNs, ViTs, and SAM leads to increased processing time and resource demands, which can limit real-time deployment in resource-constrained environments such as edge computing platforms or mobile agricultural robots. To address this, optimizing the model architecture through knowledge distillation, pruning, or lightweight transformer variants could significantly reduce computational overhead while maintaining segmentation precision. Furthermore, the generalizability of the model across diverse crop types and field conditions remains an area for further investigation. Although the current framework demonstrates strong performance on densely fruited crops, variations in environmental factors, lighting conditions, and crop structures may introduce challenges when applied to different agricultural datasets. Future research could leverage domain adaptation strategies or self-supervised learning techniques to enhance model adaptability without requiring extensive manual annotations.

4.3. Future research directions

In future work, we will address the issues related to occlusion and clustering by incorporating statistical refinements, such as boundary-aware object estimation, to better handle occlusions and clustering. The integration of synthetic data generation and self-supervised learning within the proposed framework could further bridge domain gaps, enhancing robustness in agricultural detection and analysis. Future research on domain adaptation and contrastive learning can improve the generalizability of our proposed model in diverse fields and crops, reducing the dependence on extensive labeled datasets. Expanding the LLM capabilities by incorporating multimodal agricultural data, including environmental sensors and weather analytics, can enable more precise predictive decision-making for yield optimization and sustainable farming strategies. Future studies can also explore collaborative AI–human frameworks where LLM-driven recommendations dynamically assist farmers in real-time decision-making, optimizing labor management, resource allocation, and risk mitigation in large-scale agricultural operations. These advancements will further expand the potential of autonomous crop monitoring, yield forecasting, and intelligent farm management.

5. Conclusion

We propose SAMConvFormer+LLM, a novel framework that integrates SAM, a Joint Convolutional Transformer, and LLMs to advance dense crop segmentation and AI-driven decision-making in precision agriculture. Using CNNs for local features, ViTs for global context, and SAM for segmentation refinement, our approach achieves precise boundary delineation, particularly in overlapping clusters and occluded regions. The LLM-driven pipeline also allows automated yield estimation, ripeness analysis, and actionable recommendations for optimized

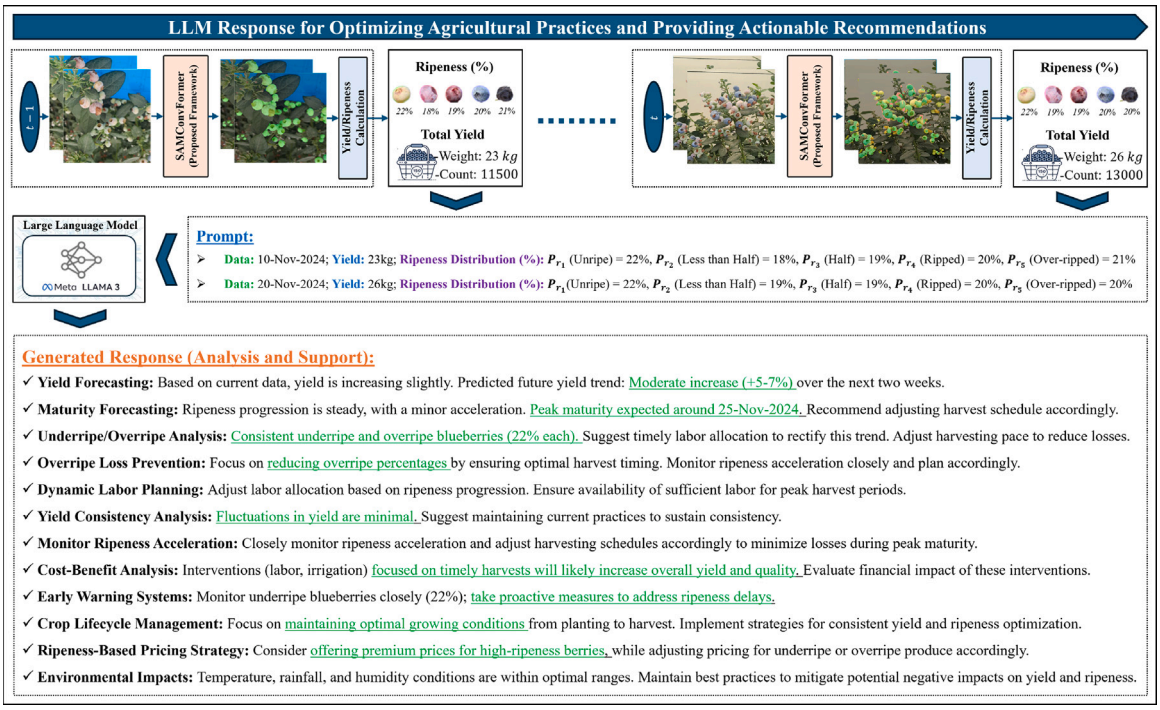


Fig. 12. Visualization of the output from the proposed LLM-driven framework, designed to optimize agricultural practices and provide actionable recommendations for improved yield, resource efficiency, and sustainable farming.

resource management and sustainable farming. Extensive evaluations demonstrate state-of-the-art performance, surpassing existing methods. The results demonstrate the ability of the model to analyze both current and historical crop data, leveraging temporal trends in yield and ripeness to provide informed agricultural insights. Beyond accuracy improvements, SAMConvFormer+LLM establishes a scalable AI-driven pipeline that bridges vision-based crop detection with intelligent decision support systems. To encourage further research in agricultural AI, we will publicly release our framework that provides a solid foundation for advancements in automated crop monitoring, yield forecasting, and precision farming solutions.

CRediT authorship contribution statement

Muhammad Owais: Writing – original draft, Methodology, Data curation, Conceptualization. **Israa Fahmy:** Validation. **Lakmal Seneviratne:** Supervision, Resources, Funding acquisition. **Irfan Hussain:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Chat GPT 4o in order to enhance language clarity and correct minor errors. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset is publicly available at figshare.

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